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ITA0609

Exp-9 : Compare Linear and Polynomial Regression using Python

```
import numpy as np

from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures
from sklearn.metrics import r2_score

# Sample dataset
X = np.array([[1], [2], [3], [4], [5]])
y = np.array([1, 4, 9, 16, 25])

# Linear Regression
linear = LinearRegression()
linear.fit(X, y)
y_linear = linear.predict(X)

# Polynomial Regression (Degree = 2)
poly = PolynomialFeatures(degree=2)
X_poly = poly.fit_transform(X)
poly_model = LinearRegression()
poly_model.fit(X_poly, y)
y_poly = poly_model.predict(X_poly)

# Display results
print("Linear Regression Predictions:")
print(y_linear)
print("\nPolynomial Regression Predictions:")
print(y_poly)
print("\nLinear Regression R2 Score:", r2_score(y, y_linear))
print("Polynomial Regression R2 Score:", r2_score(y, y_poly))
```

```
*** Linear Regression Predictions:  
[-1.  5. 11. 17. 23.]
```

```
Polynomial Regression Predictions:  
[ 1.  4.  9. 16. 25.]
```

```
Linear Regression R2 Score: 0.9625668449197862  
Polynomial Regression R2 Score: 1.0
```